When Does a Transient Accretion Disk Form? Comparison of Wind-Fed and RLOF Accretion in Eccentric Binaries

ROHIT KUMAR & AMIT KASHI, *Ariel University*

ARIEL UNIVERSITY

2026 INTERNATIONAL IAU–ORAS PRO–AM WORKSHOP

BACKGROUND

Massive binary stars of which more than 70% of main-sequence and early-type stars exist in binary systems evolve through strong stellar winds, mass transfer, and angular momentum exchange. When a massive star (the donor) releases its wind, the gravity of the companion star (the accretor) captures a portion of that wind; this process is known as Bondi–Hoyle–Lyttleton (BHL) accretion. In close binaries, when the donor star evolves and fills its Roche lobe, material flows directly towards the accretor via the L_1 point a process called Roche Lobe Overflow (RLOF), which is another significant mechanism of mass transfer.

METHODS FOR BHL

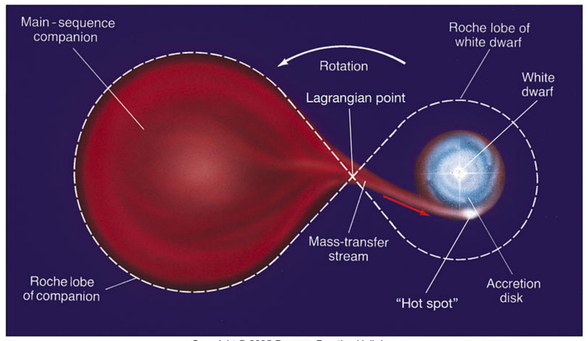
The captured gas possesses angular momentum. Whether the material falls directly onto the accretor or first forms a disk depends on whether its angular momentum is greater or less than the Keplerian value at the accretor's surface.

$$j_A = \Omega R_{\rm acc}^2, \qquad j_2 = \sqrt{G M_a R_a}.$$

Formation criteria: Disk: $\dfrac{j_A}{j_2} \geq 1$,

Belt: $0.8 \leq \dfrac{j_A}{j_2} < 1$, No Disk: $\dfrac{j_A}{j_2} < 0.8$

RLOF



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METHODS FOR RLOF

Eggleton (1983) [1] approximation evaluated across the full orbit (1° steps):

$$\frac{R_{L,d}(\theta)}{r(\theta)} = \frac{0.49\, q_d^{2/3}}{0.6\, q_d^{2/3} + \ln(1 + q_d^{1/3})}$$

Quantity	Value
Roche-lobe radius (periastron)	$189.1\, R_\odot$
Donor physical radius	$180\, R_\odot$

No classical RLOF occurs anywhere along the orbit.

BHL RESULT

Adopted parameters: $M_d = 170\, M_\odot,\ M_a = 80\, M_\odot,\ a = 19.74\,\mathrm{AU},\ e = 0.9,\ P = 5.54\,\mathrm{yr}$

Quantity	Value
Disk window	32°–114°
Thick Disk/Belt	16°–31° & 115°–118°
Duration	$\sim$ 34 days
Peak lag	$\sim$ 11 days

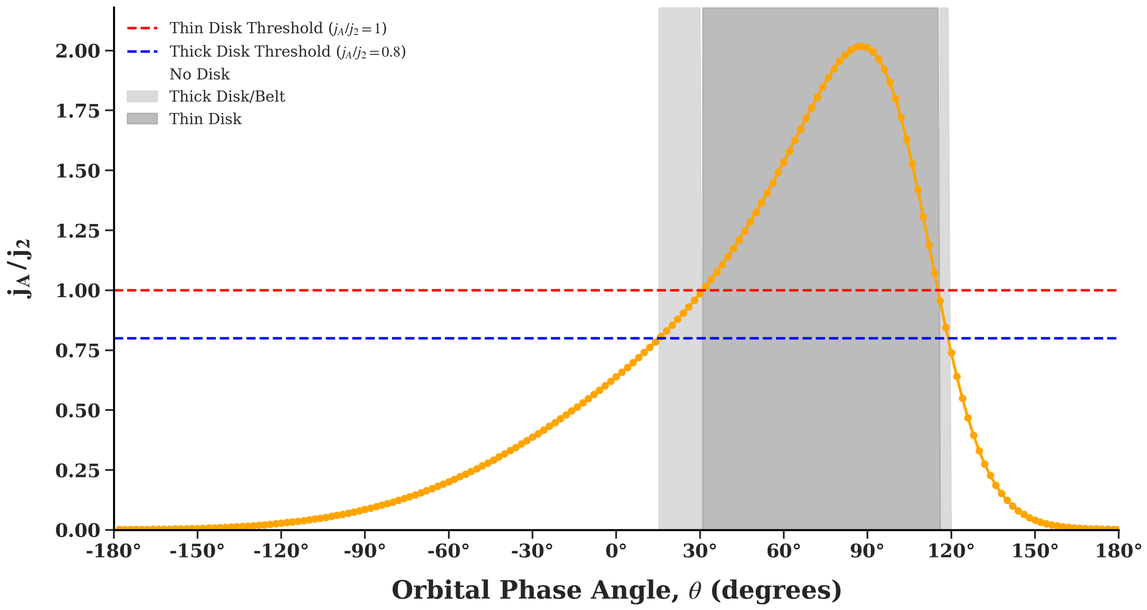


Figure 1: j_A/j_2 across orbital phase, showing disk, belt, and no-disk regions.

CONCLUSION

- The wind-fed disk forming around η Carinae's companion star is **transient**. it builds up only during a narrow orbital window (32°–114°, spanning just ∼34 days of the entire 2023-day orbit).
- From plot, the peak accretion rate occurs not at periastron, but ∼**11 days after** periastron—meaning maximum accretion does not coincide with the exact moment of closest approach.
- Disk formation requires a favourable combination of wind speed, orbital separation, and accretor size; no single parameter is sufficient on its own.
- The plot shows that j_A/j_2 increases smoothly with the orbital phase—the transition from **No Disk** to **Thick Disk/Belt** and then to **Thin Disk** is gradual, not abrupt.
- With the adopted parameters, the Roche lobe is not filled; therefore, classical RLOF **does not occur**. However, the system is **highly sensitive to eccentricity**—a change of just $\Delta e \approx 0.005$ is sufficient to trigger overflow.

OUTLOOK

- **MESA:** Combine the analytical BHL model with MESA to include self-consistent stellar evolution, winds, and changing stellar properties.
- **Multiple Orbits:** Simulate multiple periastron passages to determine whether the accretion disk/belt survives between successive passages or dissipates.
- **Parameter Study:** Vary wind velocity, β, stellar masses, mass ratio, and eccentricity to identify disk/belt formation conditions and compare predictions with systems such as η Carinae.

REFERENCES

[1] P. P. Eggleton. Approximations to the radii of roche lobes. *The Astrophysical Journal*, 1983.

[2] Andrew Frank and Derek. *Accretion Power in Astrophysics*. Cambridge University Press, 2002.

[3] Amit Kashi and Noam Soker. Possible implications of mass accretion in eta carinae. *New Astronomy*, 2009.